

Traffic Engineering vs. Content Distribution: A Game Theoretic Perspective

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Abstract—In this paper we explore the interaction between content distribution and traffic engineering. Because a traffic engineer may be unaware of the structure of content distribution systems or overlay networks, his management of the network does not fully anticipate how traffic might change as a result of his actions. Content distribution systems that assign servers at the application level can respond very rapidly to changes in the routing of the network. Consequently, the traffic engineer's decisions may almost never be applied to the intended traffic.

We use a game-theoretic framework in which infinitesimal users of a network select the source of content, and the traffic engineer decides how the traffic will route through the network. We formulate a game and prove the existence of equilibria. Additionally, we present a setting in which equilibria are socially optimal, essentially unique, and stable. Conditions under which efficiency loss may be bounded are presented, and the results are extended to the cases of general overlay networks and multiple autonomous systems.

I. INTRODUCTION

Internet Service Providers employ traffic engineering on their networks to control how resources are used and to optimize performance on their network. While protocols such as Multi Protocol Label Switching (MPLS) allow ISPs to permit sources some control over how their traffic is routed to its destination [1], presently it is primarily the role of the ISP to decide what path the traffic will take through its own network (and hence which resources the traffic will utilize).

The ISP's performance optimization may be frustrated, however, by a response to its traffic engineering by the users of the network. Typical peer-to-peer networks and content distribution systems have the property that a user is not concerned about which particular server she uses. If traffic to a server is slowed as a result of traffic engineering, she may break the connection and form a new one with a different server. If server selection is implemented by an application the user is running, these connections may respond on a much faster time scale than the ISP's traffic engineering—the ISP's policies may almost never be applied to the intended traffic. In a general overlay network, the change in traffic may be even more drastic—as traffic is routed through the overlay network, several different traffic flows in the underlying network may be created.

The authors thank the National Science Foundation and a Stanford Graduate Fellowship for support, as well as helpful discussions with Michael J. Freedman, Frank Kelly, and Damon Wischik.

This interaction has a fundamental impact on the effectiveness of traditional traffic engineering policies. In particular, we must examine whether traffic engineering is beneficial in a setting where content selection happens at the edges of the network. Under what conditions does the interaction between content distribution and traffic engineering yield efficient outcomes, and when is the interaction detrimental to overall network performance?

To answer such questions, this paper models the interaction between users' overlay routing and the ISP's traffic engineering in a game-theoretic setting. Our main contributions are the following.

- 1) We develop and rigorously analyze a model where the traffic engineer chooses a routing policy, and selfish users choose the best available content sources. We show that for a fixed routing policy of the traffic engineer, there exists a simple characterization of equilibria among infinitesimal users that generalizes the well known Wardrop equilibrium. We then show that under mild assumptions, there always exists an equilibrium of the game between the traffic engineers and the users.
- 2) We show that if the congestion signals generated by the network are properly matched to the objective of the traffic engineer, then equilibria of this game are fully efficient. Further, we show that in this case natural dynamics converge to this optimal outcome, where the traffic engineer updates his routing policy over time in response to observed traffic, and users choose content sources in response to the engineer's policy.
- 3) We show that if the congestion signals generated by the network are not properly matched to the objective of the traffic engineer, then significant inefficiencies may result; in fact, the resulting equilibrium may be worse than that obtained under pure source routing, and may not even be Pareto optimal. We further relate our game to the selfish routing model of [2], and use their techniques to show that the traffic engineer may ensure stability and bound his efficiency loss.
- 4) We conclude by extending our results to general overlay networks, and networks consisting of multiple ISPs.

Section II discusses previous literature on the topics of content distribution, overlay networks, and traffic engineering.

Section III presents a model of distribution of content to selfish users. Section IV extends these results to the more general setting of selfish users accessing an overlay network. In Section V, some observations on overlay networks and inter-domain routing are presented.

II. RELATED WORK

There has been some interest in the interaction between overlay networks and traffic engineering. Content distribution networks may be viewed as a special case of overlay networks, where the overlay routes are direct from a server to a logical node that represents the content being served.

In [3], the authors investigate the interaction between selfish routing, overlay networks, and traffic engineering. Their approach is simulation-based, and in their scenarios they find that the outcome of selfish overlay routing is similar to that of selfish source routing (principally, they find low average latency but heavy congestion at a few links).

In [4], the authors formulate a two-player game between an overlay network operator and a traffic engineer. The traffic engineer minimizes network cost, while the overlay network operator is concerned only with traffic across the overlay network. They find that due to these competing objectives, unstable dynamics and high costs may result.

The authors in [5] present a model that is used to examine the interaction between content distribution and traffic engineering. As in [4], most of their focus is on a content distributor with an objective distinct from that of the traffic engineer. They explore the Pareto curve of this two-player system, and propose a Nash bargaining solution as a fair outcome. In contrast to the settings of [5] and [4], we assume a content provider with limited control over its users' decisions, and we are principally concerned with the traffic engineer's objective.

In [6], the authors investigate the interaction between selfish routing and traffic engineering. In this model, traffic engineers instantaneously optimize their local networks as selfish users choose through which networks they wish to route their traffic. In addition to having a different model of traffic engineering, our paper assumes that traffic engineers optimize on a slower scale.

The authors in [7] explore the impact of overlay networks on ISP peering decisions. In [8], the authors explore a traffic engineering protocol based on the principles of Wardrop equilibria.

There have also been several simulation-based studies of equilibria among selfish users of overlay networks. In [9], the authors explore user dynamics in such a system. Users act in a selfish fashion but are not infinitesimal and do not balance their flows fractionally across their routes. The authors find that this leads to significant inefficiencies and instability.

Additionally, there has been some research into multipath routing, overlay networks, and TCP. In [10], the authors present conditions under which traffic engineering and TCP can coexist in a stable manner. This is similar to our setting

in that traffic engineering may affect the resulting traffic. In [11], TCP is analyzed in the context of multipath routing.

III. INTRA-DOMAIN TRAFFIC ENGINEERING AND CONTENT DISTRIBUTION

Consider a network that consists of a set of resources J together with a set of servers S and a set of content types K . The resources may be thought of as edges of the network. There is a set of users M – each user $m \in M$ has demands d_m^k for each content type $k \in K$. Servers host content – each $s \in S$ is associated with a set of content types $K_s \subseteq K$. From this we may define S_k , the set of servers associated with content k , as $\{s \in S \mid k \in K_s\}$.

For each user m and server s there is a (possibly empty) set of routes R_{ms} that user m can use to access server s . Call user m *connected* to server s if R_{ms} is nonempty. Each route $r \in R_{ms}$ is identified with the resources in J that it utilizes – i.e., each r is a nonempty subset of J , not necessarily disjoint from other routes. User m satisfies his demands for content k using the servers in S_k . We suppose moreover that this demand may be feasibly satisfied – that is, if $d_m^k > 0$ then there must be at least one server in S_k to which m is connected.

Suppose that the users do not have the ability to choose the routes themselves. Instead, there is an additional player, the *traffic engineer*, who is tasked with determining the manner in which each player routes to each server.

The predominant intra-domain routing protocols, OSPF and IS-IS, do not allow for completely arbitrary assignment of traffic to routes. In particular, an ISP's ability to split traffic among available routes is constrained. However, there is evidence that properly tuned networks employing OSPF may perform within a few percent of optimal routing with arbitrary flow splitting allowed [12], [13]. Additionally, protocols such as MPLS give the ISP additional control. Accordingly, in this paper we utilize an idealization in which the traffic engineer may arbitrarily distribute traffic from a source to a destination among the paths connecting those points.

We suppose that the engineer chooses for each (m, s) a distribution over R_{ms} whenever R_{ms} is nonempty. For $r \in R_{ms}$, let π_{mrs} be the proportion of traffic between m and s that will utilize route r .

Definition 3.1: A *routing policy* is a vector $\vec{\pi} = (\pi_{mrs})$ over $m \in M, s \in S, r \in R_{ms}$ that is a distribution over routes for each connected user-server pair; that is, for all $m \in M$ and $s \in S$ such that $R_{ms} \neq \emptyset$, there holds $\sum_{r \in R_{ms}} \pi_{mrs} = 1$.

Suppose the engineer has chosen his routing policy $\vec{\pi}$. Then the users must choose servers to satisfy their demands for the content. That is, user m chooses $e_{ms}^k \geq 0$ such that $\sum_{s \in S_k} e_{ms}^k = d_m^k$ for all k ; thus e_{ms}^k is the amount of content k traffic delivered to user m from server s . (We assume also that if m and s are not connected then $e_{ms}^k = 0$.) The traffic from server s to user m is then $e_{ms} = \sum_{k \in K_s} e_{ms}^k$. This results in flows f_j through each resource j ; given a routing policy $\vec{\pi}$ and vector of demands $\vec{e}$, we then have resulting

flows:

$$f_j(\vec{e}, \vec{\pi}) = \sum_{ms} \sum_{r: j \in r, r \in R_{ms}} \pi_{mrs} e_{ms}. \quad (1)$$

A. The Engineer's Problem

We suppose that the engineer is unaware of the existence of the content types, and naively takes the user-server traffic as given. When he chooses the distributions over routes, he does not anticipate that user demand will change between the servers. He is only able to observe the total traffic between each user-server pair – as above, e_{ms} is the quantity of traffic he believes needs to be routed between user m and server s .

We assume that associated with each resource $j \in J$ is a nonnegative, nondecreasing, convex, and differentiable function $L_j(f_j)$ representing the social cost incurred when resource j carries flow f_j . For example, this function may represent the load at resource j or the total latency experienced by users due to j . The engineer wishes to minimize the total cost of the network. His problem is to find a routing policy $\vec{\pi}$ that minimizes this over all possible distributions over routes. That is, given demands $\vec{e}$, he chooses $\vec{\pi}$ to solve the following optimization problem:

$$\begin{aligned} & \text{minimize} \quad \sum_{j \in J} L_j(f_j(\vec{e}, \vec{\pi})) \\ & \text{s.t.} \quad \pi_{mrs} \geq 0, \quad \forall r \in R_{ms}, m, s; \\ & \quad \sum_{r \in R_{ms}} \pi_{mrs} = 1, \quad \forall m, s \text{ s.t. } R_{ms} \neq \emptyset. \end{aligned} \quad (2)$$

B. The Users' Problem

As the users do not choose their routes, their only decision is which servers they will use to satisfy their demand for content. In practice, overlay networks react to a variety of signals from the network. For example, peer-to-peer clients such as BitTorrent will select other peers on the basis of the download rate delivered; thus, implicitly, a client favors downloading from other peers where the intervening paths exhibit low congestion. More generally, we consider an abstract model where each link is associated with a *price* that depends on the total flow through it; we assume that end systems react to the sum of link prices on a path. These prices can represent measured congestion or delay; more generally, they may be congestion signals generated by the network provider (e.g., using Explicit Congestion Notification); similar approaches have been taken in the literature on congestion control (see, e.g., [14], [15]). Of course, since end systems cannot choose the actual path taken by content delivered from a server, the price observed will in fact be an *average* price over all paths available.

We now make this informal discussion precise. Associated with each resource $j \in J$ is a nonnegative, nondecreasing, and continuous price function $p_j(f_j)$. The price of a route r is simply the sum of the prices of the resources used by that route. Accordingly, let $p_r(\vec{f}_r) = \sum_{j \in r} p_j(f_j)$, where $\vec{f}_r$ is the vector of flows through the resources in r . Assume the routing policy is $\vec{\pi}$. Then to each connected server s , user m

sees expected price $p_{ms}(\vec{f}) = \sum_{r \in R_{ms}} \pi_{mrs} p_r(\vec{f}_r)$. Here, $\vec{f}$ is the vector of resource flows. To simplify the analysis, if m and s are not connected, set $p_{ms}(\vec{f}) = +\infty$.

Given a routing policy $\vec{\pi}$ and fixing the responses of the other users, an infinitesimal user of type m chooses servers to minimize the prices he faces to access his content. That is, for each $k \in K$ and $s \in S_k$ with $e_{ms}^k > 0$, we must have

$$p_{ms}(\vec{f}) \leq p_{ms'}(\vec{f}) \quad (3)$$

for all $s' \in S_k$. In aggregate, these choices must satisfy demand for content as noted above – that is, $\sum_{s \in S_k} e_{ms}^k = d_m^k$ for all k .

We assume that the users will make these decisions independently and selfishly. The resulting set of decisions (the existence of which will be shown shortly) constitutes a best response for the users, and is in the style of a Wardrop equilibrium [16]. Let $\vec{e}$ denote the vector of e_{ms}^k for all connected $m \in M$, $s \in S$, $k \in K_s$.

Definition 3.2: Suppose the traffic engineer has chosen routing policy $\vec{\pi}$. Then $\vec{e}$ is a *best response for the users* if and only if for each $k \in K$ and $s \in S_k$ with $e_{ms}^k > 0$, and for all $s' \in S_k$, inequality (3) holds.

There is a simple characterization of the equilibrium that results among the users when $\vec{\pi}$ is fixed. Let $h_j(f_j) = \int_0^{f_j} p_j(t) dt$.

Proposition 3.1: Suppose the users face the routing policy $\vec{\pi}$. Then a best response for the users exists, and $\vec{e}$ is a best response for the users if and only if it is a solution to the following optimization problem:

$$\begin{aligned} & \text{minimize} \quad \sum_{j \in J} h_j(f_j(\vec{e}, \vec{\pi})) \\ & \text{s.t.} \quad \sum_{s \in S_k} e_{ms}^k = d_m^k, \quad \forall m \in M, k \in K; \\ & \quad e_{ms}^k \geq 0 \quad \forall m, s, k \in K_s; \\ & \quad e_{ms}^k = 0 \quad \forall m, s, k \in K_s, R_{ms} = \emptyset. \end{aligned} \quad (4)$$

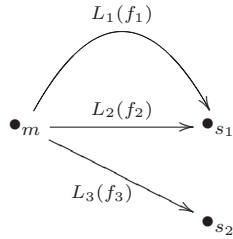
Proof: Since each $p_j(f_j)$ is nondecreasing, the function $h_j(f_j)$ is convex and differentiable. The optimality conditions of this convex minimization program are readily seen to be equivalent to the definition of a best response. Existence of a best response follows from the existence of a solution to this minimization problem. ■

The objective in (4) is the same as that of the selfish routing model in [2]. As in [2], convexity implies that this equilibrium is unique with respect to the resulting prices. Note that each user sees an *average* price to each connected server, where the average is computed using the traffic splitting policy of the engineer. In particular, this means that the expected price functions seen by two users will typically be quite different, even if an identical route is shared by them. Nevertheless, they behave as though they were in a normal Wardrop equilibrium with added restrictions on possible flows. This is somewhat surprising, because here it is not possible to cast the users' decisions as between price functions that are common to all users, except in highly symmetric cases. This characterization

applies to all routing policies, not merely those that arise as a best response by the traffic engineer.

C. Equilibrium Concept

Intuitively, we would expect the system to be in equilibrium when the traffic engineer has optimized the total latency with respect to the user-server flows, and the users are utilizing flows to content that minimize the prices they face. Unfortunately, these simple requirements admit a number of undesirable situations – examples such as the following show that these requirements allow outcomes where the resulting flows are arbitrarily bad compared to the optimal routing policy, motivating a refinement that will be introduced shortly.



In the above diagram, there is a single mass of users at node m who desire unit flow of content, which can be served by the two servers s_1 and s_2 . The set of resources J is the set of edges 1, 2, and 3. Routes are direct, and the traffic engineer's routing policy to s_2 is trivial, since there is only one route to s_2 . Hence the traffic engineer's sole decision is whether to use edge 1 or edge 2 to route traffic from s_1 .

Suppose that $L_1(f_1) = f_1$, $L_2(f_2) = (1000 + \epsilon)f_2$, and $L_3(f_3) = 1000f_3$. Further, suppose $p_j(f_j) = L'_j(f_j)$ for each edge j , so that $p_1(f_1) = 1$, $p_2(f_2) = 1000 + \epsilon$, $p_3(f_3) = 1000$. The optimal flow has cost 1 and occurs when the traffic engineer uses link 1 and the user accesses s_1 . Consider, however, the situation in which the traffic engineer's policy is to route all traffic originated by s_1 through link 2. Suppose also that the users receive traffic only from s_2 . Clearly this is a best response for the users, since they see price 1000 for s_2 , which is less than the price $1000 + \epsilon$ for content from s_1 .

Furthermore, the traffic engineer's routing policy is a solution to optimization problem (2) – indeed, any routing policy is a solution, since there is no traffic between m and s_1 . But the resulting flows have cost 1000 – orders of magnitude higher than the cost of the optimal flow.

These situations arise due to an inherent ambiguity in the traffic engineer's problem: in a static setting he does not differentiate among routing policies for user-server pairs between which there is not currently any traffic. Inspecting the optimization problem (2), it is clear that when $e_{ms} = 0$, any distribution over $(\pi_{mrs} : r \in R_{ms})$ is equivalent as far as the traffic engineer is concerned.

This motivates a refinement via a regularity condition: the traffic engineer should pick routes over these pairs that would properly handle an infinitesimal quantity of traffic, were it to occur. Specifically, if $\pi_{mrs} > 0$, we would like the marginal cost along this route to be at least as good as that along any

other route between m and s . That is, for all $m \in M$, $s \in S$ and $r \in R_{ms}$ such that $\pi_{mrs} > 0$, and all $r' \in R_{ms}$, we require:

$$\sum_{j \in r} L'_j(f_j) \leq \sum_{j \in r'} L'_j(f_j). \quad (5)$$

One can easily verify that for user-server pairs with $e_{ms} > 0$, this restriction is already implied by the Karush-Kuhn-Tucker (KKT) conditions of (2). This will enable us to compactly describe the engineer's best response as a policy such that (5) holds for all $\pi_{mrs} > 0$ and $r' \in R_{ms}$. In a slight abuse of notation, we allow $\vec{e}$ to refer also to the vector of user-server traffic e_{ms} for all connected $m \in M$, $s \in S$.

Proposition 3.2: Fix the user-server traffic $\vec{e}$. Then there exists a routing policy $\vec{\pi}$ that is optimal for (2) and satisfies inequality (5) for all $\pi_{mrs} > 0$ and all $r' \in R_{ms}$. Further, any routing policy $\vec{\pi}$ that satisfies inequality (5) for all $\pi_{mrs} > 0$ and all $r' \in R_{ms}$ is also optimal for (2).

Proof: For the first part, we have already noted that the optimality conditions of (2) imply (5) for $e_{ms} > 0$, and that for $e_{ms} = 0$ we may freely choose π_{mrs} as far as (2) is concerned. So it suffices to set $\pi_{mrs} = 1$ for some $r \in R_{ms}$ that minimizes $\sum_{j \in r} L'_j(f_j)$.

The second part follows from the KKT conditions for (2). The assumption that the elements of $\vec{\pi}$ yield probability distributions, together with (5), implies that $\vec{\pi}$ is optimal for (2). ■

Since we have shown that this refinement is not overly restrictive, we use it to define the traffic engineer's best response. Routes that receive positive probability should have lowest marginal cost, regardless of whether or not they are being utilized.

Definition 3.3: Suppose the traffic engineer faces user-server demands $\vec{e}$. Then a routing policy $\vec{\pi}$ is a *best response for the traffic engineer* if and only if for all $m \in M$, $s \in S$ and $r \in R_{ms}$ such that $\pi_{mrs} > 0$, and all $r' \in R_{ms}$, inequality (5) holds.

We say that a routing policy $\vec{\pi}$ together with user demands $\vec{e}$ is an *equilibrium* if $\vec{e}$ is a best response for the users to $\vec{\pi}$, and $\vec{\pi}$ is a best response for the traffic engineer to $\vec{e}$.

Proposition 3.3: Suppose the latency functions $L_j(f_j)$ are nonnegative, nondecreasing, convex and differentiable in f_j for all $j \in J$, and that $p_j(f_j)$ is nonnegative, nondecreasing, and continuous. Then there exists an equilibrium.

Proof: We reinterpret our model via the following convex game with two players. The first player, the engineer, chooses $\vec{\pi}$ to minimize total latency as in (2). The second player, the user, chooses $\vec{e}$ to minimize $\sum_{j \in J} h_j(f_j)$ as in (4). The players' strategy spaces are nonempty, convex, and compact, and their payoff functions are continuous and quasi-convex in these terms. So there exists a pure-strategy Nash equilibrium to this game by Nash's Theorem [17].

This equilibrium will not necessarily satisfy the refinement to the traffic engineer's response, however. We might have $e_{ms} = 0$, in which setting it is not guaranteed by the optimality conditions of (2). To work around this, we add demands to

enforce $e_{ms} > 0$. Consider a modified setting in which the system is augmented with an additional set of content types $\bar{K} = \{k_s \mid s \in S\}$. For these new content types, let $d_m^{k_s} = \epsilon \mathbb{I}_{[R_{ms} \neq \emptyset]}$ and $S_{k_s} = \{s\}$. That is, between each connected user-server pair, we add a small demand for content that can only be satisfied by traffic through that pair. For this modified setting, then, we have a Nash equilibrium as above in which $e_{ms} > 0$ for all connected user-server pairs. Consequently, this Nash equilibrium satisfies our equilibrium refinement purely from the optimality conditions.

To obtain the desired equilibrium in our original game, let $(\epsilon) \searrow 0$ be a decreasing sequence, and let $\bar{\pi}(\epsilon)$ and $\bar{e}(\epsilon)$ be a Nash equilibrium for this game with these added demands, for each ϵ in the sequence. Since these strategies are in a compact space, there exists a limit point, i.e., a limit strategy $\bar{\pi}$ and $\bar{e}$, together with an infinite subsequence of strategies converging to it, indexed by ϵ_n , $n \in \mathbb{N}$. Let $(f_j(\epsilon_n))$ be the resulting flows for each n , and let (f_j) be the flows resulting from the limit point. Clearly, $(f_j(\epsilon_n)) \rightarrow (f_j)$.

Note first that (f_j) and (e_{ms}^k) satisfy our original demands d_m^k , since they satisfy them for each point in the sequence. Next, note that $\bar{\pi}$ is a routing policy, again since each member of the sequence is a routing policy. Also, if $\pi_{mrs} > 0$, then for all sufficiently large n we have $\pi_{mrs}(\epsilon_n) > 0$ and hence, from the optimality conditions of (2), we have

$$\sum_{j \in r} L'_j(f_j(\epsilon_n)) \leq \sum_{j \in r'} L'_j(f_j(\epsilon_n)). \quad (6)$$

Taking the limit, the continuity of $L_j(f_j)$ implies that $\bar{\pi}$ satisfies refinement (5).

Similarly, if $e_{ms}^k > 0$, then for all large enough n , $e_{ms}^k(\epsilon_n) > 0$, and from the optimality conditions of (4),

$$p_{ms}(\bar{f}(\epsilon_n)) \leq p_{ms'}(\bar{f}(\epsilon_n)). \quad (7)$$

Taking the limit, by the continuity of $p_j(f_j)$, this implies (3). Hence, $\bar{\pi}$ and (e_{ms}^k) form an equilibrium. ■

D. Pigovian Taxes and Efficiency

In this section we consider a particular special case with desirable properties: suppose that $p_j(f_j) = L'_j(f_j)$, so that $h_j(f_j) = L_j(f_j)$. We will show that in this case there not only exists an equilibrium, but it is socially optimal; and further, best response dynamics between the users and the traffic engineer converge to the optimal cost. Since the objective functions of (2) and (4) are identical in this setting, this is essentially a potential game as studied in [18]. However, we utilize a continuous strategy space and employ a regularity condition for the traffic engineer, so the results of that paper do not apply directly.

As a motivating case, consider the setting of [2] in which $L_j(f_j) = f_j l_j(f_j)$ is the relevant cost function for given delay functions $l_j(f_j)$. In this setting, $l_j(f_j)$ is the queueing delay experienced by each unit of traffic using resource j – hence the total delay experienced at resource j is $f_j l_j(f_j)$. In this case $p_j(f_j) = l_j(f_j) + f_j l'_j(f_j)$; the additional term $f_j l'_j(f_j)$ beyond the delay is a *Pigovian tax* [19], which ensures that

users correctly internalize the congestion effect they have on others [20]. In this section, we consider a setting where these prices can be correctly fed back to the end user via explicit congestion signals.

Consider the following minimization, which captures what a social planner might attempt. Since $\sum_{j \in J} L_j(f_j)$ is the social cost, a well-informed planner would seek to minimize this over all possible routing policies and user-server traffic. Let g_{mrs}^k represent the flow from m to s along route r for the purpose of serving content k . If $\vec{g}$ is the vector of such flows, then the resulting flow across resource j is

$$f_j(\vec{g}) = \sum_{ms} \sum_{k \in K_s} \sum_{r: j \in r, r \in R_{ms}} g_{mrs}^k. \quad (8)$$

Then the social optimization problem may be presented as:

$$\begin{aligned} & \text{minimize} \quad \sum_{j \in J} L_j(f_j(\vec{g})) \\ & \text{s.t.} \quad \sum_{s \in S_k} \sum_{r \in R_{ms}} g_{mrs}^k = d_m^k, \quad \forall m \in M, k \in K \\ & \quad \quad g_{mrs}^k \geq 0, \quad \forall m, s, r \in R_{ms}, k \in K_s. \end{aligned} \quad (9)$$

One can readily translate between these flows and associated routing policies and user-server traffic. Suppose (g_{mrs}^k) is an optimal solution to this program, and $\vec{f} = (f_j)$ is the vector of optimal flows. Then for all m, s , and $k \in K_s$, let $\hat{e}_{ms}^k = \sum_{r \in R_{ms}} g_{mrs}^k$, the implied traffic between m and s for the purpose of serving content k .

Fix a user-server pair (m, s) with $R_{ms} \neq \emptyset$. If $g_{mrs}^k > 0$ for some $r \in R_{ms}$ and $k \in K$, then we set $\hat{\pi}_{mrs}$ to the split that is implied by g_{mrs}^k – that is, we set $\hat{\pi}_{mrs} = (\sum_{k \in K_s} g_{mrs}^k) / (\sum_{\bar{r} \in R_{ms}} \sum_{k \in K_s} g_{m\bar{r}s}^k)$. Otherwise, we have $\hat{e}_{ms} = 0$; in this case we set $\hat{\pi}_{mrs} = 1$ for some r that minimizes $\sum_{j \in r} L'_j(f_j)$.

It is worth noting that the resulting routing policy and $(\hat{e}_{ms}^k)$ imply the optimal flows $\vec{f}$. The proof, omitted, follows from an examination of the feasible sets of the respective convex programs.

Proposition 3.4: The derived policy $(\hat{\pi}_{mrs})$ and traffic $(\hat{e}_{ms}^k)$ form an equilibrium.

Conversely, there is a correspondence from equilibria to solutions of this program.

Proposition 3.5: Suppose $\bar{\pi}$ and $\bar{e}$ form an equilibrium, yielding flows $\vec{f}$. Then these flows together with $g_{mrs}^k = \pi_{mrs} e_{ms}^k$ are an optimal solution to (9).

Proof: It is easily checked that this derived (g_{mrs}^k) is feasible. To verify optimality, it suffices to show that the KKT conditions hold. This optimality condition is that whenever $g_{mrs}^k > 0$, we have for all $s' \in S_k$ and $r' \in R_{ms'}$

$$\sum_{j \in r} L'_j(f_j) \leq \sum_{j \in r'} L'_j(f_j). \quad (10)$$

By our choice of $p_j(f_j)$, this inequality can be rewritten as $p_r(\vec{f}_r) \leq p_{r'}(\vec{f}_{r'})$.

Suppose $g_{mrs}^k > 0$ for some $m \in M, s \in S_k, r \in R_{ms}$. Fix $s' \in S_k$ and $r' \in R_{ms'}$. From our definition of g_{mrs}^k , we know that $e_{ms}^k > 0$ and $\pi_{mrs} > 0$.

Let $\mu_{ms} = \min_{\vec{r} \in R_{ms}} p_{\vec{r}}(\vec{f}_{\vec{r}})$. Similarly, set $\mu_{ms'} = \min_{\vec{r} \in R_{ms'}} p_{\vec{r}}(\vec{f}_{\vec{r}})$. Since $\vec{\pi}$ is a best response for the traffic engineer, we know that whenever $\pi_{mrs} > 0$ we must have $p_{\vec{r}}(\vec{f}_{\vec{r}}) = \mu_{ms}$. In particular, since $\pi_{mrs} > 0$, we have $p_r(\vec{f}_r) = \mu_{ms}$.

We have $\mu_{ms} = \sum_{\vec{r} \in R_{ms}} \pi_{mrs} \mu_{ms} = \sum_{\vec{r} \in R_{ms}} \pi_{mrs} p_{\vec{r}}(\vec{f}_{\vec{r}})$, since we have term-wise equality when $\pi_{mrs} > 0$ as above. Hence $p_r(\vec{f}_r) = \mu_{ms} = p_{ms}(\vec{f})$.

Since $\vec{e}$ is a best response for the users and $e_{ms}^k > 0$, it follows that $p_{ms}(\vec{f}) \leq p_{ms'}(\vec{f})$. But by a similar argument as above, $p_{ms'}(\vec{f}) = \mu_{ms'}$. Since $\mu_{ms'} \leq p_{r'}(\vec{f}_{r'})$ by construction, we conclude that $p_r(\vec{f}_r) \leq p_{r'}(\vec{f}_{r'})$, and these flows are optimal for (9). ■

E. Pigovian Taxes and Dynamics

From the perspective of users of a system, a traffic engineer updates his policy very infrequently. Our dynamics intend to capture the notion that between moves by the traffic engineer, the users have enough time to converge to an equilibrium among themselves. These two time scales are captured by dynamics consisting of alternating moves between the traffic engineer and the collective of users. Similar processes are studied in [5] and [4]. We have shown that there is essentially only one equilibrium outcome as far as social cost is concerned. We can use this fact, together with the commonality of the players' objective functions, to show that dynamics converge to this outcome.

Consider an arbitrary initial routing policy $\vec{\pi}(0)$. At odd times t , let $\vec{e}(t)$ be a best response of the users to routing policy $\vec{\pi}(t-1)$. At even times, let $\vec{\pi}(t)$ be a best response of the traffic engineer to user demands $\vec{e}(t-1)$. Let $\vec{f}(t)$ be the flows at time t , and let $C(t) = \sum_{j \in J} L_j(f_j(t))$. Let C be the socially optimal cost, as in (9).

Proposition 3.6: These best-response dynamics have cost that approaches the social optimum. That is, $\lim_{t \rightarrow \infty} C(t) = C$.

Proof: Note that $C(t)$ is a nonincreasing sequence, since it is the common objective function of both the users' and engineer's respective optimization problems, and can only be improved by a best response. Furthermore, it is bounded below by C . Hence it converges to some limit point $\hat{C} \geq C$. To show equality, it suffices to show that it is the cost of some equilibrium, since Proposition 3.5 establishes that all equilibria have cost C .

A natural candidate is the pair of policies that result from a limit point. Consider the joint sequence $(\vec{e}(t), \vec{\pi}(t+1))$ for odd t . These are user demands that arise in best-response dynamics, together with a best response for the traffic engineer.

This joint vector is restricted to a compact set, so the sequence has a limit point $(\vec{e}, \vec{\pi})$, together with an infinite subsequence $(\vec{e}(t_n), \vec{\pi}(t_n+1))$ that converges to it for some increasing $t_n \uparrow \infty$. Continuity arguments similar to those used in Proposition 3.3 show that $\vec{\pi}$ is a routing policy, that $\vec{e}$ satisfies users' demands, that together these have cost $\hat{C}$, and that $\vec{\pi}$ is a best response for the traffic engineer to $\vec{e}$.

All that needs to be established is that $\vec{e}$ is a best response for the users to $\vec{\pi}$. Suppose otherwise for the sake of contradiction. Then there is some other response $(\vec{e}_{ms}^k)$ that results in cost $\bar{C} < \hat{C}$. But by continuity of the users' objective function, for large enough n there is an ϵ such that the user demands $(\vec{e}_{ms}^k)$ will result in cost $\bar{C} + \epsilon < \hat{C}$ when faced with routing policy $\vec{\pi}(t_n + 1)$. This implies that $C(t_n + 2) < \hat{C}$, a contradiction since this is a decreasing sequence with limit point $\hat{C}$.

Hence these limit policies $\vec{e}$ and $\vec{\pi}$ form an equilibrium, and $\hat{C} = C$. ■

While the costs must converge in the limit, due to the plurality of the players' best responses the strategies may not. However, in some circumstances the resulting flows will be unique. In particular, if $L_j(f_j)$ is strictly convex for each $j \in J$, then the optimal flows will be unique, and hence the dynamics will converge to these flows.

F. General Prices and Efficiency Loss

In contrast to subsection III-D, here we consider the case where the user prices do not necessarily lead to a socially optimal result. Though Proposition 3.3 implies that an equilibrium exists for more general prices than those required by Proposition 3.5, this equilibrium may have undesirable properties. It may be suboptimal, dynamics need not converge, and the equilibrium may be worse than a related selfish routing outcome. Indeed, we will find that when the prices do not reflect the correct Pigovian taxes, the equilibrium outcome may be even worse than one where all routing control is ceded to users (i.e., full source routing). The equilibrium may also not be Pareto optimal, in the sense that both the social objective and the users' fictitious objective may be underserved. Nonetheless, by considering analogues of the bounds on price of anarchy found in [2], convergence of dynamics and bounds on efficiency loss may be ensured in special cases.

In this section we consider the setting where $L_j(f_j) = f_j l_j(f_j)$, and focus on the case where $p_j(f_j) = l_j(f_j)$ for all $j \in J$. The traffic engineer's objective is to minimize total latency, but users respond to measured latency without regard for the externality they impose on others. This is a well-studied case in which bounds on the efficiency loss of selfish routing may be established [2].

While the socially optimal problem minimizes $\sum_{j \in J} f_j l_j(f_j)$, which is also the traffic engineer's objective, the users will attempt to minimize $\sum_{j \in J} \int_0^{f_j} l_j(z) dz$. These objectives are not aligned, and there is no reason to believe that a solution to the latter will minimize the former.

Let $\vec{f}^{OPT}$ denote the socially optimal set of flows, realized by routing policy $\vec{\pi}^{OPT}$ and user demands $\vec{e}^{OPT}$. To measure cost of the players' choices, let $C(\vec{\pi}, \vec{e}) = \sum_{j \in J} f_j l_j(f_j)$, where $f_j = \sum_{ms} \sum_{r \in R_{ms}} \pi_{mrs} \sum_{k \in K} e_{ms}^k$, be the cost when the traffic routing policy is $\vec{\pi}$ and the users choose $\vec{e}$.

A useful reference point is the set of flows that would result if the users had full source routing in this setting of selfish infinitesimal users. Let $h_j(x) = \int_0^x l_j(z) dz$. Consider

the following optimization problem, the solution to which is the set of equilibrium flows, as established in [16].

$$\begin{aligned} & \text{minimize} \quad \sum_{j \in J} h_j(f_j(\vec{g})) \\ & \text{s.t.} \quad \sum_{s \in S_k} \sum_{r \in R_{ms}} g_{mrs}^k = d_m^k, \quad \forall m \in M, k \in K \\ & \quad g_{mrs}^k \geq 0 \quad \forall m, s, r \in R_{ms}, k \in K_s. \end{aligned} \quad (11)$$

Let $\vec{f}^{WE}$ be the vector of flows resulting from a solution to (11). As we did with the socially optimal flows, we can derive from these a routing policy $\vec{\pi}^{WE}$ and user demands $\vec{e}^{WE}$.

Indeed, in examples in which the traffic engineer's problem (and hence his control) is trivial – that is, each R_{ms} is empty or a singleton – the problem reduces to selfish routing. These examples conform to the efficiency bounds established in [2]. For example, in the case where $|R_{ms}| \in \{0, 1\}$ and each $l_j(f_j)$ is affine, we have $C(\vec{\pi}^{WE}, \vec{e}^{WE}) \leq \frac{4}{3}C(\vec{\pi}^{OPT}, \vec{e}^{OPT})$.

If the traffic engineer is concerned with maintaining this sort of bound, there are ways he can meet this goal.

Corollary 3.7: Suppose instead of minimizing total delay, the traffic engineer chooses a routing policy to solve

$$\begin{aligned} & \text{minimize} \quad \sum_{j \in J} h_j(f_j(\vec{e}, \vec{\pi})) \\ & \text{s.t.} \quad \pi_{mrs} \geq 0, \quad \forall r \in R_{ms}, m, s; \\ & \quad \sum_{r \in R_{ms}} \pi_{mrs} = 1, \quad \forall m, s \text{ s.t. } R_{ms} \neq \emptyset. \end{aligned} \quad (12)$$

That is, the traffic engineer aligns his objective with that of the users. Additionally suppose that the traffic engineer adheres to a refinement analogous to Definition 3.3, so that when $e_{ms} = 0$ he only assigns positive probability to routes that minimize the latency $\sum_{j \in r} l_j(f_j)$. Then the cost resulting from best-response dynamics converges to $C(\vec{\pi}^{WE}, \vec{e}^{WE})$.

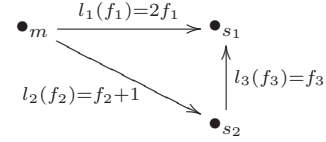
This follows from the same argument we used to show convergence to $C(\vec{\pi}^{OPT}, \vec{e}^{OPT})$ in the setting of Proposition 3.6, since the objective functions of the traffic engineer, the users, and the selfish routing problem are identical. Though straightforward, this is an interesting result: it offers the traffic engineer a way to effectively grant the users source routing privileges (by mimicking the outcome that would result with pure source routing), even when protocols do not otherwise permit them.

Another efficiency bound can be established, this time in terms of the inefficiency induced by the users' best response. In particular, suppose that the traffic engineer is fully informed of the entire setup – how many content types there are, which servers supply them, and how much demand for each the users have. In this case the engineer can compute the optimal policy $\vec{\pi}^{OPT}$; however, the users' best response to this policy may be suboptimal. The following proposition bounds the effect of this inefficiency. The proof, omitted, utilizes the variational inequality technique employed in [21].

Proposition 3.8: Suppose the traffic engineer uses routing policy $\vec{\pi}^{OPT}$. Let $(\hat{e}_{ms}^k)$ be a best response by the users. Sup-

pose also that each $l_j(f_j)$ is affine. Then $C(\vec{\pi}^{OPT}, (\hat{e}_{ms}^k)) \leq \frac{4}{3}C(\vec{\pi}^{OPT}, \vec{e}^{OPT})$.

We conclude with an important observation: that in general, an equilibrium may be *less* efficient than the outcome achieved by full source routing. Since this discussion aims to provide similar bounds on cost as achieved by source routing, it is natural to ask whether one can guarantee that an equilibrium does at least as well as $C(\vec{\pi}^{WE}, \vec{e}^{WE})$. The answer is no, as the following example demonstrates.



In the above diagram, there is a single user at node m who desires unit flow of content, which can be served by the two servers s_1 and s_2 . The set of resources J is the set of edges 1, 2, and 3, which are labeled by their latencies. There is one route to s_2 , direct through edge 2. There are two routes to s_1 – traffic may proceed either directly across edge 1, or be routed via edges 2 and 3. Hence the traffic engineer's sole decision is to determine what proportion of traffic destined for s_1 proceeds there directly.

Clearly, in both the socially optimal and selfish routing idealizations, no traffic would ever travel to s_1 through the indirect route. In this example, $C(\vec{\pi}^{OPT}, \vec{e}^{OPT}) = 1.25$ and $C(\vec{\pi}^{WE}, \vec{e}^{WE}) = 1.33$. In equilibrium, however, the traffic engineer will route some traffic indirectly, and this will be sufficiently detrimental to yield cost 1.39, greater than in selfish routing. This example also demonstrates that the equilibrium need not be Pareto optimal – here, both the users' and traffic engineer's objective functions would be strictly improved if the flows were $\vec{f}^{WE}$.

This is an important insight: it shows that when the congestion signals are not correctly chosen, the resulting equilibrium might actually leave both users and the engineer worse off than a simple “hands off” approach of allowing pure source-based route selection. Indeed, it is worth noting that in the case where the price signals include the Pigovian taxes (cf. Section III-D), the Wardrop equilibrium (i.e., the full source routing solution) would also minimize the total social cost; however, today's networks do not generally allow full end system control over routes. Thus, in light of our negative results in this section, the results of Section III-D may be reinterpreted as suggesting that *traffic engineering can serve to efficiently guide end users if the congestion signals are chosen correctly*.

IV. GENERAL OVERLAY NETWORKS

In this section we shift from the special case of content distribution to the more general setting of overlay networks, where most of the above results will carry over. As an application of this generalization, we show that one can analyze a setting where servers themselves have access links that can become congested, and are not under control of the traffic engineer.

In an overlay network, traffic flows from a source to a destination on a sequence of logical links. Each logical link is actually an origin-destination pair in the physical network, and is routed according to the traffic engineer's routing policy. Thus while users may have source routing on the overlay network, the physical route that their traffic takes is in part constrained by the traffic engineer's policy. We assume that selfish users conduct overlay source routing based on measured prices that are a result of the physical traffic.

As before, our physical network consists of a set of resources J , a set of origins M and a set of destinations S , together with routes R_{ms} for $m \in M$ and $s \in S$ that are identified with the resources consumed along those routes. The traffic engineer's problem will remain the same – he sees demands e_{ms} from origins to destinations, and chooses a routing policy $\vec{\pi}$ that determines on which routes that traffic will flow.

Associated with the physical network is a logical network, or overlay network. It consists of a set of logical users M^O and servers S^O , together with a set of routes $\mathcal{R}$. Each user-server pair $u \in M^O, v \in S^O$ is associated with a set of routes $R_{uv}^O \subseteq \mathcal{R}$.

Each user-server pair wishes to route quantity d_{uv}^O of traffic, and selfishly assigns this traffic to the available routes. Denote by $e_{u\sigma v}^O$ the quantity of traffic u sends to v along route $\sigma \in R_{uv}^O$, so that $\sum_{\sigma \in R_{uv}^O} e_{u\sigma v}^O = d_{uv}^O$. For notational convenience, assume that when $\sigma \notin R_{uv}^O$ we fix $e_{u\sigma v}^O = 0$.

Let $\vec{e}^O$ be the vector of all $e_{u\sigma v}^O$. We assume that a route in the logical network is a set of origin-destination pairs in the physical network. That is, for $m \in M, s \in S$, if $(m, s) \in \sigma$ then traffic along route σ generates traffic between m and s in the physical network. Then for $\sigma \in \mathcal{R}$, the total flow along this route is

$$f_{\sigma}^O(\vec{e}^O) = \sum_{u \in M^O, v \in S^O} e_{u\sigma v}^O \quad (13)$$

The total traffic demand generated in the physical network between m and s is $e_{ms} = \sum_{\sigma \in \mathcal{R}: (m,s) \in \sigma} f_{\sigma}^O$. As before, this is the demand that the traffic engineer will optimize against.

Let $\vec{f} = (f_j)$ be the flows that result in the physical network as a result of the overlay traffic and the routing policy, as in (1). We assume again that users are infinitesimal and selfish. As in Section III, we assume that the price functions are on links, and so the expected price from physical origin $m \in M$ to destination $s \in S$ is $p_{ms}(\vec{f})$ as defined in Subsection III-B. Then the expected price of route $\sigma \in \mathcal{R}$ is $p_{\sigma}(\vec{f}) = \sum_{(m,s) \in \sigma} p_{ms}(\vec{f})$.

It is now straightforward to extend the results of this paper to this more general setting. The users should only use routes of minimum expected price; i.e., if $e_{u\sigma v}^O > 0$ then for any $\phi \in R_{uv}^O$ we must have $p_{\sigma}(\vec{f}) \leq p_{\phi}(\vec{f})$. For a fixed routing policy $\vec{\pi}$, we can analogously characterize the equilibrium among the users, as in Proposition 3.1. On the other hand, the traffic engineer faces the same problem as in Section III, so it is simple to define an equilibrium concept that incorporates the overlay users; and an analogue of Proposition

3.3 then establishes existence of an equilibrium, as long as the latency functions $l_j(f_j)$ are nonnegative, nondecreasing, convex and differentiable in f_j for all $j \in J$, and each $p_j(f_j)$ is nonnegative, nondecreasing, and continuous. The remaining results of Sections III-D and III-F carry over in a transparent manner; in particular, if the correct Pigovian taxes are reflected in the prices, then the equilibrium is efficient and best response dynamics converge, while if the prices are not correctly aligned with the overall objective, the resulting outcome may be more inefficient than pure source routing.

We conclude this section by recalling that (as remarked in Section I) content distribution is a special case of an overlay network. This allows us to generalize the content distribution model of the paper; for example, suppose that we consider content distribution networks that have congested access links at the servers. This can be reflected by simply by adding additional physical resources over which the traffic engineer has only trivial routing control. These nodes may then be connected to the content via logical connections in the overlay network. More generally, it is possible to account for server-side resource congestion specific to all traffic at a specific server or, alternatively, dependent upon the quantity served of a particular type of content.

V. MULTIPLE ISPS

The results in sections III and IV translate to a situation with multiple ISPs performing intra-domain network optimization, provided that the users are in control of inter-domain routing decisions. In the absence of this restriction on inter-domain routing decisions, individuals ISPs are capable of inflicting virtually unlimited damage on downstream ISPs for selfish reasons. We consider a case where ISPs are connected to each other via logical nodes on the boundaries of the networks; in particular, we assume that for any two ISPs, there exists a unique logical node connecting them. In this situation, inter-domain routing decisions will implicitly be made on the basis of the overlay routing decisions made by users. The class of networks we consider includes those in [6], which also presents examples that elucidate why this assumption is needed. The social objective is to minimize the total latency for all resources, across all domains.

To make the model more precise, suppose there is a set of domains $\mathcal{N}$. Each domain $n \in \mathcal{N}$ is concerned with a set of resources J_n , and domains have disjoint resource sets. Origin-destination pairs are restricted to the same domain (e.g. M_n and S_n), and the routes between them utilize only resources from that domain. Thus all inter-domain "traffic" is the result of the overlay network – specifically, it results from overlay routes that contain origin-destination pairs from distinct domains. Conceptually, this is due to the location of logical nodes on the boundaries of the networks.

Each domain n chooses as before a routing policy $\vec{\pi}^n$ for those origin-destination pairs that reside in his domain. That is, he chooses $\vec{\pi}^n$ to solve the following optimization problem,

given (e_{ms}) for $m \in M_n, s \in S_n$:

$$\begin{aligned} & \text{minimize} && \sum_{j \in J_n} L_j(f_j(\vec{e}, \vec{\pi})) \\ & \text{s.t.} && \pi_{mrs} \geq 0 \quad \forall r \in R_{ms}, m \in M_n, s \in S_n; \\ & && \sum_{r \in R_{ms}} \pi_{mrs} = 1 \\ & && \forall m \in M_n, s \in S_n \text{ s.t. } R_{ms} \neq \emptyset. \end{aligned} \quad (14)$$

It is a simple observation that our assumptions result in completely independent optimization problems for the individual traffic engineers, given fixed overlay traffic. Consequently they may be treated as a single traffic engineer for the purpose of classifying equilibria, and the earlier static results apply in this setting. The following proposition is established by combining this observation with Proposition 3.4.

Proposition 5.1: Let $\mathcal{N} = \{1, \dots, N\}$. Consider an arbitrary initial routing policy $\vec{\pi}^n(0)$ for each $n \in \mathcal{N}$. Suppose that at odd times t , the users choose $\vec{e}(t)$ to be a best response to the routing policies $\vec{\pi}^n(t-1)$. Suppose that at even times, each ISP n chooses independently with probability $p_n > 0$ to select and update their routing policy, so that $\vec{\pi}^n(t)$ is a best response to $\vec{e}(t-1)$; otherwise (with probability $1-p_n$) $\vec{\pi}^n(t) = \vec{\pi}^n(t-1)$. Then the resulting dynamics have costs that converge almost surely to the optimal cost.

Proof: Let A_t denote the set of ISPs that update their policies at time t . Observe that almost surely, $A_t = \mathcal{N}$ infinitely often. Restricting our attention to these subsequences (together with the preceding moves by the users) and looking at a limit, the result follows pointwise as in the proof of Proposition 3.4, upon noting that when all the ISPs update it is as if there were a single ISP. ■

This implies that properly motivated users are potentially better at choosing socially optimal inter-domain routes than are traffic engineers. This insight provides another argument for the utility of projects such as Resilient Overlay Networks [22]. However, this optimality would not hold in practice, as a substantial quantity of inter-domain traffic is unrelated to overlay networks. Additionally, there is an incentive for autonomous systems to engage in price inflation, in which they exaggerate their congestion in order to shift overlay traffic elsewhere.

VI. CONCLUSION

In this paper we have provided insights into the nature of selfish routing on an overlay network and how it interacts with traffic engineering on the underlying network. Neither stability nor efficiency is to be expected in general. In some cases, however, efficiency loss may be bounded or eliminated.

With the proper signals, the fictitious objective of the users may be aligned with that of the traffic engineer. From a design standpoint, if applications on overlay networks greedily make connections based on these signals, then the interaction between traffic engineering and selfish routing will lead to an optimal distribution of traffic across the network. This

interaction would be fruitful even if the overlay network were distributed across several independent domains, provided that users determine their inter-domain routing.

On the other hand, without the proper signals, significant inefficiencies can result in the interactions between users and ISPs. We outline several methods of determining this efficiency loss; a principal insight is that the interaction of traffic engineering and content distribution may be even worse than the outcome achieved with pure source routing in this setting.

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